

Global Horizon Bank

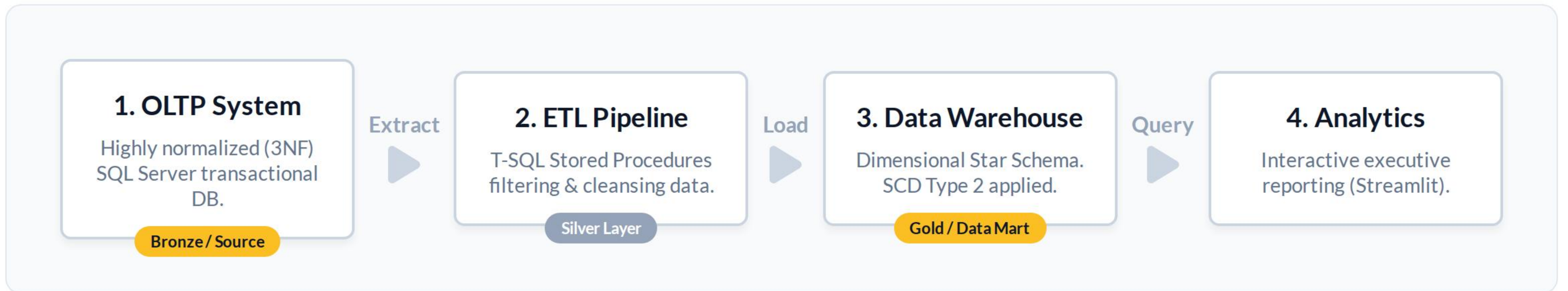
Data Architecture

End-to-End Enterprise Data Pipeline (OLTP to OLAP)

Applying Medallion Architecture & Data Lineage

Enterprise Data Pipeline Overview

The high-level data flow representing the extraction, transformation, and loading of banking data into analytical structures.

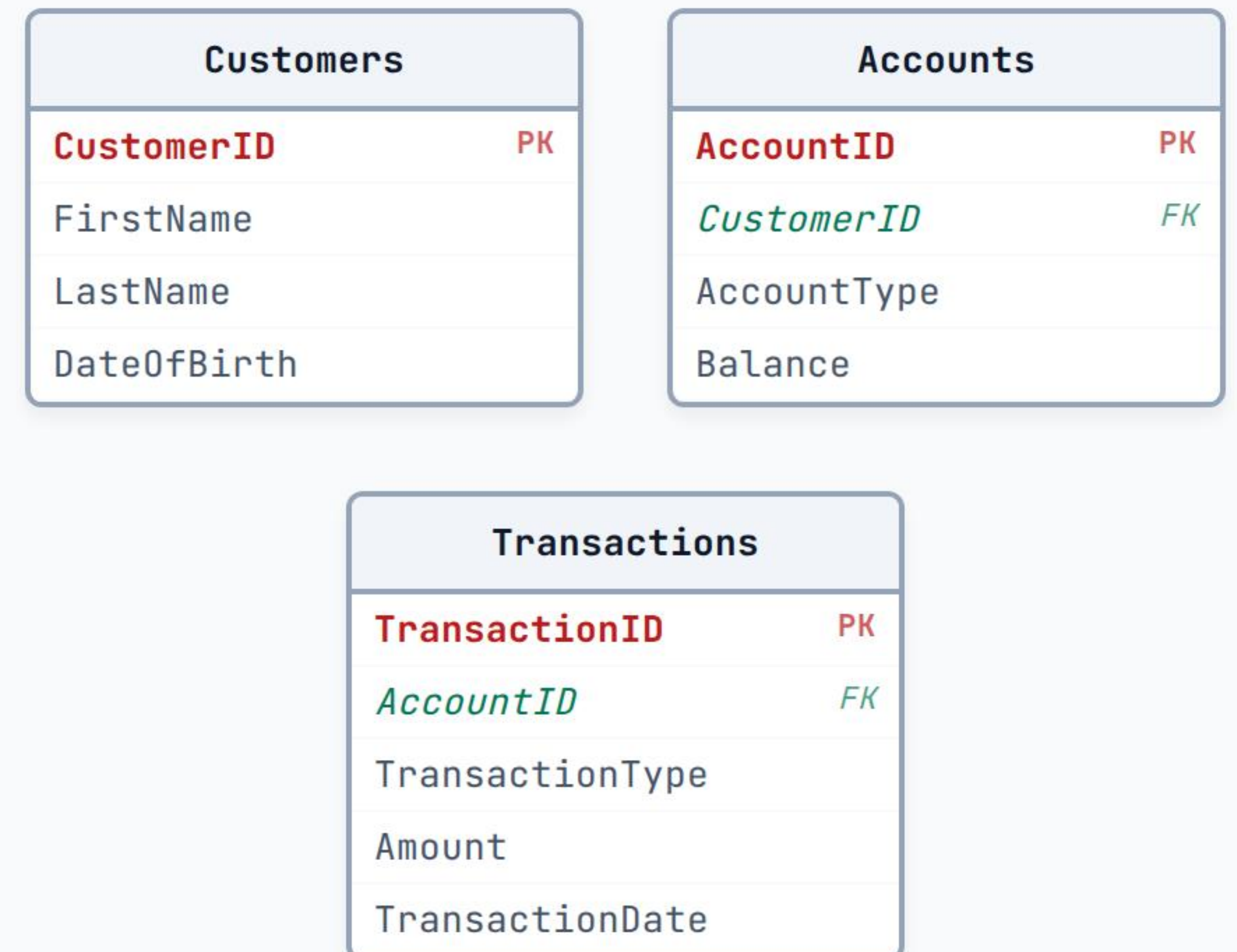


Phase 1: Source to Bronze (OLTP Schema)

Extracting Raw Transactions

Data is ingested from the core operational database as-is. The structure is highly normalized (3NF) to support transactional integrity, not analytical querying.

-  **Relational Complexity:** Requires complex joins across Customers, Accounts, and Transactions.
-  **Raw State:** Contains raw strings and operational IDs without historical tracking.



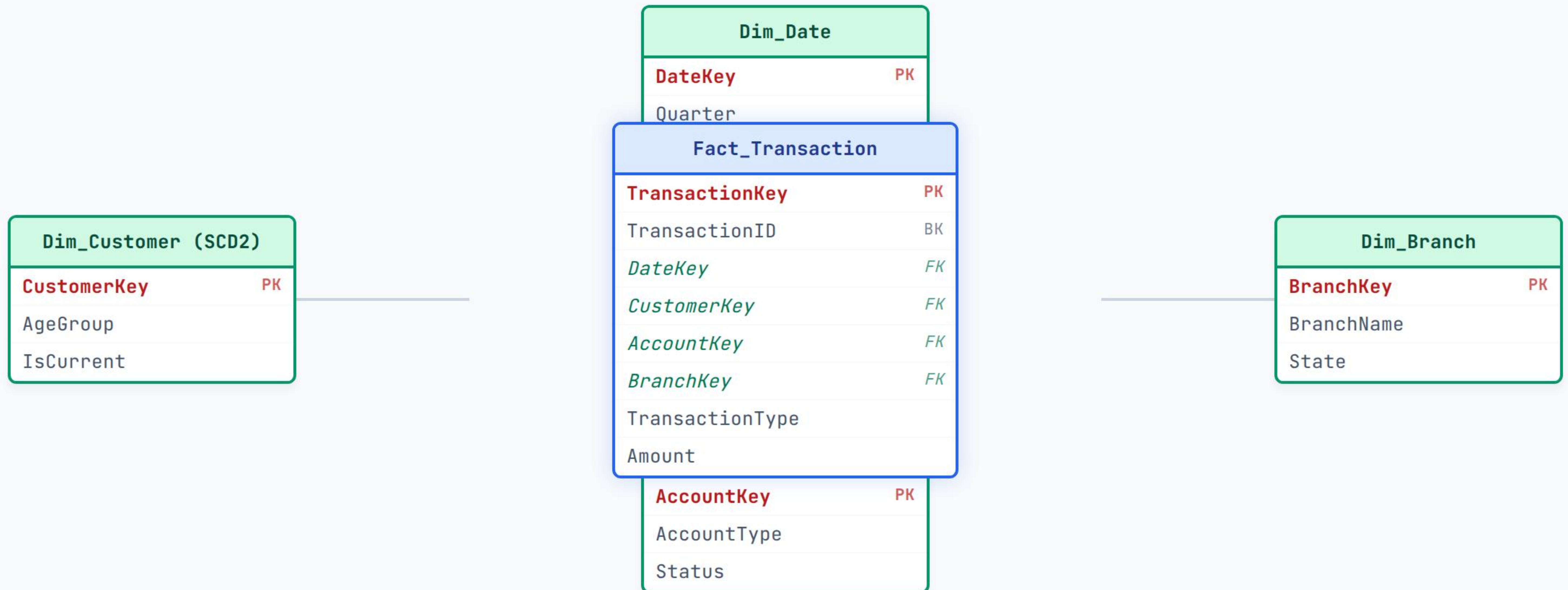
Data Lineage: Transformation in Silver

Mapping the evolution of attributes from the normalized OLTP source to the denormalized OLAP targets through the ETL process.

OLTP Source Field (Bronze)	ETL Transformation Logic (Silver)	Data Mart Target (Gold)
Customers.CustomerID	→ Generate Surrogate Key, Apply SCD2 logic	Dim_Customer.CustomerKey (PK)
Transactions.TransactionDate	→ Parse, Extract temporal components (Qtr, Day)	Dim_Date.DateKey (FK)
Transactions.Amount	→ Cleanse nulls, cast to fixed decimal	Fact_Transaction.Amount
Customers.DateOfBirth	→ Calculate age dynamically, bucket into ranges	Dim_Customer.AgeGroup

Phase 3: The Gold Layer (OLAP Star Schema)

The heavily transformed data is exposed as a Star Schema, optimizing read performance and analytical queries for the dashboard.



Pipeline Complete

Data structure is successfully optimized for dashboard ingestion.